

Liam Shannon

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EDUCATION

Queen's University

Expected May 2028

BASc in Mechatronics and Robotics Engineering | Dean's Scholar | GPA: 4.19/4.3

EXPERIENCE

Perception Manager

July 2026 - Present

Queen's Aerospace Design Team

Kingston, Canada

- Developing a Computer Vision stack by implementing physical AI world models and training a YOLO object-detection model to identify specified task ground targets from a ZED 3D depth-sensing camera mid-flight
- Directing a 15-member team building the software stack in C++ and Python for a fixed-wing autonomous drone to compete at the advanced class of the SAE Aero 2027 design competition
- Managing a GitHub organization and a Dockerized Ubuntu 24.04 build system enabling consistent cross-platform development across macOS, Windows, and Linux for all contributors

Software Developer

June 2026 - September 2026

Cavtera

Ottawa, Canada

- Saved \$1,000,000+ annually in recoverable costs through exposing 10,000+ hours per week of idling equipment by building a full-stack equipment management system using TypeScript and React
- Increased job-site utilization of 1,000+ trucks and pieces of equipment by shipping automated alerts
- Restored live equipment telematics by fixing broken ingestion APIs and rebuilding the Python transform pipeline that parses 10,000+ records per day across a database of millions
- Modeled the business as a Foundry ontology and wrote the TypeScript edit functions behind it, giving non-technical staff a governed way to update production records
- Shipped the operator-facing front end as a React and TypeScript application on Foundry's Ontology SDK (OSDK), so site staff could act on live fleet data directly

Software Developer

May 2025 - April 2026

Thomas Cavanagh Construction Limited

Ottawa, Canada / Remote

- Digitized dispatch for a 100+ driver fleet by building custom dispatch software
- Eliminated 4+ hours of daily dispatcher calls and dozens of spreadsheets by modeling fleet data in an ontology
- Reduced compute costs from \$100+/week to single digits by caching data locally instead of querying the ontology
- Delivered the dispatcher-facing tooling as Foundry Workshop applications backed by Python transforms

PROJECTS

Autonomous Mars Rover Robot | ROS, Raspberry Pi 4, Arduino, SLAM

January 2026 - April 2026

- Designed a fully autonomous four-wheeled rover using ROS on a Raspberry Pi 4 for decision-making, sensor processing, and simultaneous localization and mapping
- Built a low-level Arduino control system handling motor drivers, encoder feedback, and PID control for precise motion

Line Following Robot | Arduino, C++, Closed-Loop Control

January 2025 - April 2025

- Developed a line-following robot using an Arduino to program motor actions based on closed-loop sensor feedback
- Tuned algorithms based on sensor data and optimized performance by integrating hardware and software subsystems

SKILLS

Languages: TypeScript, JavaScript, Python, C/C++

Tools & Frameworks: ROS2, Claude Code, Ontology building, React, Git, Docker, Linux

Data: Python data pipelines, data cleaning & normalization, SQL-style querying/Foundry querying logic

Other: SolidWorks, Onshape, Altium, LTspice